

Quantum-Assisted Clinical Risk Factor Selection for Diabetes Readmission Prediction

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Abstract Hospital readmission in patients with diabetes is a relevant clinical outcome associated with disease complexity, care continuity, and post-discharge management. Early identification of patients at risk of 30-day readmission can support targeted follow-up and preventive interventions. However, clinical datasets often include heterogeneous, redundant, and noisy variables, making feature selection important for compact and interpretable predictive modelling. We propose a QUBO/QAOA-based framework for clinical risk factor selection in diabetes readmission prediction. Candidate clinical variables are encoded as binary decision variables, and feature selection is formulated as a Quadratic Unconstrained Binary Optimization problem that balances predictive relevance, redundancy reduction, and subset compactness. The resulting QUBO is mapped to an Ising Hamiltonian and optimized using shallow Quantum Approximate Optimization Algorithm circuits executed on a quantum processing unit. We evaluated the framework on the UCI Diabetes 130-US Hospitals dataset for 30-day readmission prediction. Selected feature subsets were assessed through logistic regression and compared with mutual information selection, LASSO-based selection, simulated annealing on the same QUBO objective, and the full candidate feature set. For $k = 10$, QAOA-QPU selected a compact subset of clinically meaningful risk factors and achieved an AUC of 0.661, compared with 0.689 obtained using all 40 candidate variables. The selected subset also reduced redundancy compared with non-QUBO baselines, although simulated annealing remained the strongest reduced-subset optimizer. These results provide an initial feasibility assessment of quantum-assisted clinical feature selection rather than evidence of quantum advantage.

1 Introduction

Hospital readmission is a relevant outcome in chronic disease management, as it may reflect patient frailty, disease complexity, limited care continuity, or unmet post-discharge needs. In diabetes care, early readmission is particularly important because diabetes is a multifactorial chronic condition, often associated with comorbidities, medication adjustments, recurrent complications, and repeated interactions with healthcare services. Identifying patients at risk of readmission within 30 days after discharge can support targeted follow-up and prevention-oriented interventions.

Machine learning methods have been increasingly investigated for 30-day readmission prediction in diabetic patients, to improve risk stratification and care planning [1, 2]. The UCI Diabetes 130-US Hospitals dataset is a widely used benchmark in this context. It contains ten years of clinical care data, from 1999 to 2008, collected from 130 US hospitals and integrated delivery networks, with more than 50 features describing patient encounters, diagnoses, medications, laboratory tests, and hospital outcomes [3]. The dataset supports the task of determining whether a diabetic patient is readmitted within 30 days after discharge [3].

Clinical prediction, however, should not be evaluated only through classification performance. In medical informatics, predictive models should also be compact, interpretable, and robust, particularly when they are intended to support prevention-oriented decisions. Clinical datasets often include heterogeneous variables, such as demographic descriptors, admission and discharge information, laboratory results, medication-related attributes, and diagnosis codes. Some variables may be noisy, redundant, weakly informative, or difficult to interpret. Feature selection can therefore reduce dimensionality, limit overfitting, support generalization, and identify concise sets of clinically meaningful risk factors [4].

Feature selection can be formulated as a combinatorial optimization problem, where each candidate variable is either selected or discarded. Quadratic Unconstrained Binary Optimization (QUBO) provides a suitable formulation for this setting by encoding feature relevance, pairwise redundancy penalties, and subset compactness within a single unconstrained binary objective [5]. This formulation is useful when the goal is to select a compact subset that retains predictive information while limiting overlap among selected variables.

QUBO problems can be mapped to Ising Hamiltonians and addressed using quantum optimization algorithms. Among them, the Quantum Approximate Optimization Algorithm (QAOA) is a variational gate-based algorithm for approximate combinatorial optimization [6]. Recent work has applied QAOA to feature selection by formulating the selection task as a QUBO problem and using sampled bitstrings to train downstream classifiers [7]. Related quantum-assisted approaches, including quantum annealing, have also been investigated for feature selection in medical settings, showing the potential of these methods while also reflecting the limits of current noisy quantum devices [8].

In this paper, we propose a QUBO/QAOA-based framework for clinical risk factor selection in diabetes readmission prediction. The aim is not to replace clinical prediction models with quantum circuits, but to use QAOA as an optimization layer for selecting compact, non-redundant, and interpretable subsets of clinical variables. The selected feature subsets are evaluated through classical predictive models and compared with mutual information selection, LASSO-based selection, simulated annealing applied to the same QUBO formulation, and the full candidate feature set.

2 Data and Methods

We used the UCI Diabetes 130-US Hospitals dataset [3], which includes ten years of clinical care data collected from 130 US hospitals and integrated delivery networks. The dataset contains more than 50 variables describing patient encounters, diagnoses, medications, laboratory tests, and readmission outcomes [3]. We addressed 30-day hospital readmission prediction. The original readmission variable separates patients readmitted within 30 days, patients readmitted after more than 30 days, and patients not readmitted. We formulated the task as binary classification, where the positive class includes patients readmitted within 30 days after discharge and the negative class includes the remaining cases.

The dataset includes numerical and categorical variables related to demographics, admission and discharge, previous healthcare utilization, diagnoses, laboratory tests, medication changes,

and diabetes treatment. The preprocessing pipeline excluded non-informative identifiers and variables unavailable at prediction time, handled missing or unknown values according to variable type, and grouped diagnosis codes into broader clinical categories to reduce sparsity and improve interpretability. It encoded categorical variables and standardized numerical variables when required by the downstream predictive model. All preprocessing steps were fitted within each training split and applied to the corresponding test split.

Feature selection can reduce dimensionality, remove noisy or redundant variables, improve generalization, and support interpretability in clinical risk prediction [4]. We constructed a candidate pool of $N = 40$ clinical variables before the quantum optimization stage. This pool was defined to fit the target quantum hardware while retaining clinically meaningful information. It includes variables related to previous inpatient, outpatient, and emergency encounters, time in hospital, number of medications, number of diagnoses, laboratory results, insulin use, diabetes medication use, medication changes, admission characteristics, discharge disposition, and diagnosis macro-categories.

We formulated clinical feature selection as a Quadratic Unconstrained Binary Optimization problem. QUBO models encode combinatorial optimization problems through binary decision variables and are widely used in quantum and quantum-inspired optimization [5]. Given N candidate clinical variables, each feature i is associated with a binary variable $x_i \in \{0, 1\}$, where $x_i = 1$ indicates that the feature is selected and $x_i = 0$ indicates that it is excluded. We defined the objective as:

$$\min_{\mathbf{x} \in \{0,1\}^N} \left[-\alpha \sum_{i=1}^N r_i x_i + \beta \sum_{i < j} c_{ij} x_i x_j + \gamma \left(\sum_{i=1}^N x_i - k \right)^2 \right], \quad (1)$$

where

$$x_i = \begin{cases} 1, & \text{if the } i\text{-th feature is selected,} \\ 0, & \text{otherwise.} \end{cases}$$

In Eq. (1), r_i measures how informative feature i is for 30-day readmission, c_{ij} measures the redundancy between features i and j , and k denotes the target number of selected features. The relevance term rewards informative clinical variables, the redundancy term penalizes overlapping predictors, and the cardinality term favours subsets close to k . We set $\alpha = 1.0$, $\beta = 0.5$, and $\gamma = 2.0$. This setting prioritizes predictive relevance, penalizes redundancy without letting it dominate relevance, and enforces the cardinality constraint strongly enough to obtain compact subsets close to the target size.

We mapped the resulting QUBO to an equivalent Ising Hamiltonian and optimized it with the Quantum Approximate Optimization Algorithm. QAOA is a variational gate-based quantum algorithm for approximate combinatorial optimization [6]. Prior work has used QAOA for feature selection by formulating the selection task as a QUBO problem and using sampled bitstrings to train downstream classifiers [7]. Quantum-assisted feature selection has also been explored in medical settings, including quantum annealing-based approaches, with results that indicate both potential and current hardware limitations [8]. In this work, we executed shallow QAOA circuits on quantum hardware and used the sampled bitstrings to select candidate clinical feature subsets. We compared QAOA-QPU with the full candidate feature set, mutual information top- k selection, LASSO-based selection, and simulated annealing applied to the same QUBO formulation.

We evaluated selected feature subsets by training classical predictive models for 30-day readmission prediction. We performed feature selection only within each training split and applied the selected subset to the corresponding test split to avoid information leakage. We

Table 1: QAOA-QPU clinical feature subset selected for $k = 10$.

Domain	Selected feature	Clinical interpretation
Prior utilization	number_inpatient	Previous hospital use
Prior utilization	number_emergency	Recent acute care access
Hospital stay	time_in_hospital	Hospitalization complexity
Treatment burden	num_medications	Medication burden
Clinical complexity	number_diagnoses	Comorbidity burden
Laboratory testing	AlCresult	Glycaemic control assessment
Diabetes treatment	insulin	Treatment intensity
Therapy management	change	Medication change during admission
Discharge context	discharge_disposition	Post-discharge care needs
Diagnosis group	diag_1_circulatory	Main circulatory diagnosis

assessed predictive performance using AUC, Recall 30-day, F1-score, and balanced accuracy. Recall 30-day denotes the recall of the positive class, namely the proportion of patients readmitted within 30 days that the model correctly identifies. We also evaluated feature subsets in terms of compactness, average pairwise redundancy, and stability across validation splits.

3 Results

3.1 Selected clinical risk factors

We evaluated the proposed framework on a candidate pool of $N = 40$ clinical variables and considered target subset sizes of $k = 5$, $k = 10$, and $k = 15$. The selected subsets included variables related to prior healthcare utilization, hospitalization complexity, medication burden, diabetes treatment, laboratory results, and discharge context. These categories capture complementary dimensions of short-term readmission risk, including patient frailty, clinical complexity, and continuity of care.

Table 1 reports the QAOA-QPU subset selected for $k = 10$. The selected variables cover prior hospital use, acute care access, length of stay, medication burden, glycaemic control assessment, therapy modification, and discharge disposition. This subset combines factors related to previous healthcare use, current hospitalization severity, treatment management, and post-discharge care needs.

3.2 Readmission prediction performance across subset sizes

We evaluated predictive performance by training logistic regression models on the selected feature subsets for 30-day hospital readmission prediction. Table 2 reports the results across subset sizes. We compared QAOA-QPU with the full candidate feature set, mutual information top- k selection, LASSO-based selection, and simulated annealing applied to the same QUBO formulation.

Performance increased with the number of selected variables, with smaller gains from $k = 10$ to $k = 15$. For $k = 10$, QAOA-QPU achieved an AUC of 0.661, a Recall 30-day of 0.403, an F1-score of 0.256, and a balanced accuracy of 0.604. Simulated annealing obtained the strongest performance among the reduced subsets, while QAOA-QPU remained competitive with mutual information and LASSO-based selection.

The full 40-feature model achieved the highest AUC, but it used all candidate variables. The QUBO/QAOA strategy therefore provides a compact feature selection alternative, with a moderate reduction in predictive performance and improved interpretability.

Table 2: Readmission prediction performance across subset sizes. Recall 30-day denotes recall for the positive class. Higher values are better for all metrics.

Method	k	AUC	Recall 30-day	F1-score	Bal. Acc.
All features	40	0.689	0.436	0.272	0.628
MI	5	0.631	0.361	0.235	0.579
MI	10	0.652	0.389	0.250	0.596
MI	15	0.664	0.404	0.259	0.606
LASSO	5	0.641	0.374	0.242	0.587
LASSO	10	0.664	0.407	0.259	0.607
LASSO	15	0.671	0.416	0.265	0.613
SA-QUBO	5	0.653	0.394	0.253	0.600
SA-QUBO	10	0.678	0.426	0.270	0.620
SA-QUBO	15	0.683	0.431	0.274	0.624
QAOA-QPU	5	0.642	0.377	0.244	0.589
QAOA-QPU	10	0.661	0.403	0.256	0.604
QAOA-QPU	15	0.668	0.412	0.263	0.611

Table 3: Feature subset quality for $k = 10$. Lower redundancy and higher stability are preferred.

Method	Avg. redundancy	Stability
MI	0.331	0.47
LASSO	0.286	0.51
SA-QUBO	0.203	0.62
QAOA-QPU	0.241	0.55

3.3 Redundancy and stability analysis

We also assessed the selected feature subsets in terms of average pairwise redundancy and stability across validation splits. This analysis complements predictive performance because clinical feature selection should favour compact, robust, and interpretable subsets.

Table 3 reports the subset-quality analysis for $k = 10$. QUBO-based methods reduced average pairwise redundancy compared with mutual information selection. This result is expected because filter-based ranking methods score variables independently and do not penalize overlap among selected predictors. For $k = 10$, QAOA-QPU obtained an average pairwise redundancy of 0.241, compared with 0.331 for mutual information and 0.286 for LASSO. Simulated annealing achieved the lowest redundancy and the highest stability, while QAOA-QPU selected a less redundant and more stable subset than the non-QUBO baselines.

These results indicate that the QUBO formulation supports compact feature selection by balancing predictive relevance, redundancy reduction, and cardinality control. Classical solvers remain strong optimization baselines, but QAOA-QPU provides a feasible quantum-assisted alternative for clinical risk factor selection.

3.4 Hardware feasibility

The 40-variable QUBO instance was executed on a 54-qubit superconducting QPU using shallow QAOA circuits. The problem size was selected to match the available quantum hardware while retaining a clinically meaningful feature selection task. The QPU was used as an optimization layer to sample candidate bitstrings, which were then decoded into feature subsets and evaluated through downstream classical classifiers.

We did not use exact state-vector simulation of the full 40-qubit QAOA circuit as a routine

baseline because of its exponential memory cost. An n -qubit state vector requires 2^n complex amplitudes, which makes exact simulation of 40-qubit circuits impractical on standard computing resources. The evaluation therefore focuses on classical feature selection baselines, simulated annealing on the same QUBO objective, QAOA-QPU execution, and downstream predictive performance.

4 Conclusion

This paper presented a QUBO/QAOA-based framework for clinical risk factor selection in 30-day diabetes readmission prediction. The proposed approach uses QAOA as a feature selection layer rather than as a replacement for classical clinical prediction models. Candidate clinical variables were encoded as binary decision variables, and the selection problem was formulated as a QUBO objective balancing predictive relevance, redundancy reduction, and subset compactness.

The experimental results show that QAOA-QPU can select compact and interpretable feature subsets with competitive predictive performance. For $k = 10$, the QAOA-QPU subset achieved an AUC of 0.661, while reducing the feature set from 40 variables to 10 clinically meaningful risk factors. The selected subset included variables related to prior healthcare utilization, hospitalization complexity, medication burden, diabetes treatment, laboratory assessment, and discharge context. Redundancy and stability analyses further indicate that the QUBO formulation can reduce overlap among selected variables compared with non-QUBO baselines.

The study should be interpreted as an initial feasibility assessment rather than evidence of quantum advantage. Classical optimization remains a strong baseline, with simulated annealing achieving the best performance among the reduced subsets. Still, the results indicate that QAOA executed on quantum hardware can provide a feasible quantum-assisted route for clinical feature selection under current hardware constraints.

Future work will investigate deeper QAOA circuits, larger and external clinical cohorts, fairness-aware and cost-sensitive QUBO formulations, and integration with calibrated clinical prediction models. Prospective validation will also be needed to assess whether the selected risk factor signatures can support discharge planning, follow-up prioritization, and continuity-of-care interventions in real-world settings.

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